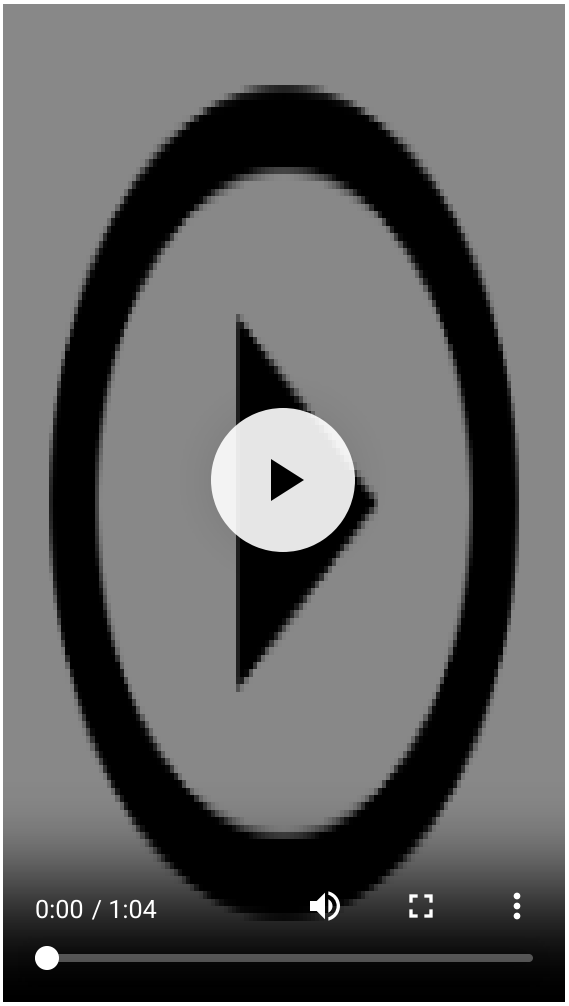
Beginner Guide: Understanding System Development and Integration.

Beginner Guide: Understanding System Development and Integration



Study the key concepts and complete the guided practice before moving to Activity 4.1.

Key Concepts

Concept	Simple meaning	Example

Development environment	Tools used to develop the system	Flutter SDK and Visual Studio
Frontend	The part of an application users interact with	Appointment request screen
Backend	Services that support application functions and data	Supabase
Database	Stores and organises information	Customer and Appointment tables
Authentication	Confirms who the user is	Signing in with an email and password
Authorisation	Determines what a user is allowed to do	Only staff can approve appointments
CRUD	Create, Read, Update and Delete operations	Add, view, change or delete an appointment
Integration	Connecting components so they work together	Flutter and ASP.NET access the same Supabase database
API	A way for applications to request data or services	Flutter requests appointment data from Supabase
Version control	Records changes to project files and code	Git
GitHub repository	A shared location for code and documentation	Group project repository
Commit	A saved record of changes	Committing the completed appointment form
Branch	A separate line of development	A branch for the login feature
Prototype	A working version that demonstrates the proposed solution	An integrated mobile and web application

Understanding the Integrated System

Your project consists of three main components.

Component	Main responsibility
Flutter	Provides the mobile application functions
ASP.NET	Provides the web application functions
Supabase	Provides the shared database and other required backend services

Short Example

A local salon wants customers to request appointments using a Flutter mobile application.

When a customer submits a request, Flutter sends the appointment information to Supabase.

Salon staff use an ASP.NET web application to retrieve the appointment information from the same database. Staff can then approve or reject the request.

When the customer opens the Flutter application again, the updated appointment status is retrieved from Supabase.

This demonstrates how the mobile and web applications work together through a shared backend.

Understanding CRUD Operations

Most information-management systems need four basic database operations.

Operation	Simple meaning	Salon example
Create	Add new information	Submit an appointment request
Read	Retrieve information	View appointment details
Update	Change existing information	Approve an appointment
Delete	Remove information	Delete an appointment where permitted

Important: Not every user should be allowed to perform every operation. Your system must follow the approved user roles and access requirements.

Guided Practice

Read the following scenario:

A local salon wants customers to request appointments using a Flutter mobile application. Salon staff will use an ASP.NET web application to manage the requests.

Both applications will use the same Supabase backend.

The group has completed its architecture diagram, use case diagram, sequence diagrams, ERD and interface designs. The group is now ready to start development.

Complete the table.

Question	**Your answer**
Which application will customers use?	
Which application will salon staff use?	
What is the purpose of Supabase?	
What happens when a customer submits an appointment request?	
Which CRUD operation is used to save a new appointment?	
Which CRUD operation is used when staff view appointment requests?	
Which CRUD operation is used when staff approve an appointment?	
How will the customer see the updated appointment status?	
Why must the two applications use the same backend?	
Where should the group maintain its source code?	
How can members provide evidence of their development contributions?	

Suggested Answer

Question	**Suggested answer**
Customer application	Flutter mobile application
Staff application	ASP.NET web application

Purpose of Supabase	Stores and manages information shared by both applications
Submitting an appointment	Flutter sends the appointment information to Supabase
Saving an appointment	Create
Viewing appointments	Read
Approving an appointment	Update
Viewing the updated status	Flutter retrieves the updated information from Supabase
Purpose of a shared backend	Both applications work with the same information
Source code location	The group's GitHub repository
Contribution evidence	GitHub commits, completed tasks and Project Progress Tracker records

Important Development Practices

During development:

- Follow the approved requirements and system designs.
- Use the same Supabase project for both applications.
- Use appropriate authentication and user access.
- Use suitable dummy data when checking functions.
- Do not upload passwords, private keys or confidential information to GitHub.
- Commit meaningful code changes regularly.
- Check that each function works before moving to the next one.
- Record actual progress and individual contributions.

Formal manual testing and final corrections will be covered in Unit 5.

Before Moving On

Make sure you can:

- explain the difference between frontend and backend;
- explain the purpose of Supabase;
- identify the four CRUD operations;
- distinguish authentication from authorisation;
- explain how Flutter and ASP.NET share information;
- explain the purpose of GitHub during development; and
- describe why development must follow the approved system designs.

You are now ready to continue to Activity 4.1: Set Up and Implement the Shared Supabase Backend.